

Titanic Survival Analysis – Project Insights

Dataset Overview

- **Analyzed 891 passenger records** from the Titanic Dataset
- Overall **survival rate** was **38.4%**, with **342 survivors** and **549 casualties**.
- Performed **data cleaning, missing value treatment, feature engineering, exploratory analysis,** and **dashboard development**.

Passenger Demographics

- Male passengers represented the majority of travelers (**577**), while females accounted for **314** passengers.
- Third Class had the highest passenger count at **491**, followed by First Class (**216**) and Second Class (**184**).
- The **21–30** age group was the largest, indicating a predominantly young adult passenger population.

Survival by Gender

- **Female survival rate** was **74.2%** compared to only **18.9%** for males
- **Gender** emerged as the **single strongest predictor** of **survival outcomes**.
- The "**women and children first**" evacuation policy is clearly reflected in the data.

Survival by Passenger Class

- **First Class passengers** recorded the **highest survival rate** of **63.0%**.
- **Second Class passengers** achieved **47.3% survival**.
- **Third Class passengers** experienced the **lowest survival rate** at **24.2%**.
- **Higher socioeconomic status** significantly improved survival chances.

Age Group Analysis

- **Children (0–12)** showed the **highest survival rate** at **58%** among all age groups.
- **Adult passengers** represented the **largest share of both survivors and casualties**.
- **Senior passengers (60+)** recorded the **lowest survival outcomes** at **30%**.

Family Size Analysis

- **Small families (2–4 members)** achieved the **highest survival rate** at **58%**.
- **Solo travellers** survived at a considerably **lower rate** of **30%**.
- **Large families (5+ members)** recorded the **poorest survival outcomes** at **16%**.

Passenger Title Analysis

- **Mrs passengers** showed the **highest survival rate** at **80%**, followed by **Miss** at **70%**.
- **Master (boys)** recorded **58% survival** reflecting **child evacuation priority**.
- **Mr passengers** experienced the **lowest survival rate** at **16%**.
- Titles provided additional insight into gender, age and social standing.

Embarkation Port Analysis

- **Cherbourg passengers** recorded the **highest survival rate** at **55.36%**.
- **Queenstown passengers** achieved **38.96% survival**.
- **Southampton** had the largest boarding volume but the **lowest survival rate** at **33.90%**.
- Survival differences across ports reflected variations in passenger class distribution.

Fare Group Analysis

- **Survival rates** increased consistently with **higher fare groups**.
- Passengers paying **fares above £101** achieved the **highest survival rate** at **74%**.
- Passengers in the **lowest fare group (0–25)** survived at only **29%**.
- Fare demonstrated a strong positive relationship with survival probability.

Dashboard Insights

- Interactive Power BI dashboard developed with thematic Titanic visual storytelling.
- Covers passenger demographics, class distribution, survival trends, family structures, fare groups and embarkation patterns
- Custom icons, matrix heatmap and themed background enhance insight exploration and user experience.

Overall Findings

- Survival was primarily influenced by gender, passenger class, age, family structure, and fare paid.
- Female passengers, children, first-class travelers, and higher-fare passengers had significantly better survival outcomes.
- Evacuation priorities and socioeconomic factors both played critical roles in determining who survived the Titanic disaster.
- The analysis demonstrates how demographic and behavioral variables together create a complete picture of survival probability